

SANTA BARBARA BUOY 46054, CA, USA

WMO: 996470

Lat: 34.265N

Lon: 120.477W

Elev: 0

StdP: 101.33

Time Zone: -8.00 (NAP)

Period: 01-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	9.1	9.7	N/A	N/A	N/A	N/A	N/A	N/A	16.3	10.5	15.7	10.6	8.2	320	0.760

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10	2.1	20.3	N/A	19.3	N/A	18.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	5.0	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
15.4	14.6	13.8	7.8	22.2	1.5	1.5	6.8	23.3	5.9	24.2	5.1	25.0	4.0	26.2		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	13.7	13.3	12.6	12.4	11.9	12.3	13.2	14.4	14.8	15.3	15.6	14.8	13.3
	DBStd	1.95	1.80	1.55	1.58	1.14	1.13	1.16	1.50	1.38	1.78	1.86	1.57	1.53
	HDD10.0	2	1	1	1	1	0	0	0	0	0	0	0	0
	HDD18.3	1709	156	159	185	194	187	154	123	111	93	87	106	155
	CDD10.0	1343	103	74	74	56	72	97	136	148	160	175	145	103
	CDD18.3	8	0	0	0	0	0	0	1	1	3	3	1	0
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	7.6	6.4	7.0	7.7	8.2	8.7	8.5	8.0	8.5	7.9	6.8	6.3	6.5
Precipitation	PrecAvg	422	91	93	68	29	11	2	1	0	2	20	33	71
	PrecMax	828	415	398	320	109	62	17	7	2	11	101	114	236
	PrecMin	158	5	3	1	3	1	0	0	0	0	1	4	6
	PrecStd	167	90	84	72	28	14	5	2	1	3	24	28	55
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.4	17.8	17.8	16.6	18.0	18.1	20.1	19.4	21.7	21.8	20.2	18.0
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	17.1	16.2	16.2	14.7	15.2	16.1	18.5	18.0	20.4	20.1	18.7	17.1
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	16.3	15.4	15.2	13.8	14.5	15.3	17.6	17.5	19.1	19.2	17.8	16.4
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	15.6	14.7	14.4	13.3	13.9	14.9	16.6	16.9	17.8	18.5	17.0	15.4
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	2.3	2.1	2.1	1.8	1.7	1.8	1.8	1.8	2.0	2.1	2.3	2.0
		MCDBR	3.5	2.9	3.4	2.6	2.6	2.7	2.2	2.0	2.8	3.1	3.6	2.4
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub	0.306	0.315	0.332	0.352	0.361	0.347	0.370	0.351	0.347	0.349	0.312	0.312	
	taud	2.611	2.599	2.546	2.470	2.469	2.506	2.496	2.514	2.536	2.528	2.605	2.604	
	Ebn at Noon	913	940	946	934	924	932	908	923	916	887	898	884	
	Edh at Noon	76	85	97	110	111	107	107	104	97	90	76	73	
All-Sky Solar Radiation	RadAvg	3.07	3.92	5.26	6.51	6.99	6.95	6.50	6.07	5.42	4.30	3.39	2.75	
	RadStd	0.23	0.37	0.33	0.39	0.55	0.38	0.31	0.37	0.37	0.39	0.20	0.23	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (24 neighbors)	+0.60	N/A	-1.49	+0.73	N/A	N/A	N/A	N/A	-136	+225	+76

Nomenclature: See separate page